

Research Papers

Safe bidirectional pulse heating method for the lithium-ion battery pack on a high-power electric motorcycle

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ABSTRACT

The charging ability at low temperatures affects the convenience and safety of electric vehicles in winter. Unlike electric cars, electric motorcycles generally do not carry high-efficiency external heating device such as heat pumps owing to strict limits on their weight and volume. This study focuses on the applicability of bidirectional pulse heating to improving the low temperature charging ability of a high-power electric motorcycle, which is also the first vehicle-used bidirectional pulse heating case. A simplified thermoelectrical model of the battery pack is proposed for onboard calculation, and a reference electrode is used to determine conservative boundary values for the bidirectional pulse and fast charging current to protect the battery cells from lithium plating and ensure the safety of the motorcycle. A map is established indicating the best preheating temperature, based on which the heating rate of the battery pack can reach 6.8 °C/min at 0 °C. The bidirectional pulse heating is tested on the electric motorcycle during winter season in Beijing, and the experimental results show that the temperature of the battery pack increases as expected. The proposed method will be applied on the motorcycle and improve with battery aging data in the future.

1. Introduction

Lithium-ion batteries (LIBs) have become the technology of choice for electric vehicles (EVs), but the range and charging time are still common concerns, especially for northern areas with cold weather [1]. Similar to other EVs, electric motorcycles suffer from performance degradation and a slow charging speed in cold weather because low temperature slows down the charge transfer and reaction in electrodes, which increases polarization. The risk of capacity loss and other safety issues is further increased by the lithium plating phenomenon, which can occur during charging at low temperatures. For motorcycles, low temperatures are defined as 0 °C or less as riders cannot be exposed to sub-zero weather for a long time. In addition, travel safety in sub-zero weather cannot be guaranteed because of the decline in tire performance and icy roads increasing the risk of slipping.

LIBs can be warmed in two ways [2,3]: external heating and internal heating. External heating uses an extra device or material to heat the battery, such as positive temperature coefficient (PTC) device, heat pump, or phase change material (PCM). Internal heating heats the LIB

with its own resistance, including heating by an excitation current and self-heating from embedding a thin nickel foil as an internal cell heater [4]. Electric motorcycles have strict limits on their weight and volume, so the energy density of the battery pack should be as high as possible, which makes internal heating more advantageous than external heating. Heating by excitation current, which is provided by the charger, is the most practical approach for electric motorcycle at present. Excitation currents can be classified as alternating current (AC) and direct current (DC) [5,6]. Bidirectional pulse can be one of the AC. Durability tests [5] have shown that heating with bidirectional pulses results in little decay of the capacity under certain conditions. Thus, it is a promising preheating method for EVs.

However, the bidirectional pulse heating is still under testing on cell or small module level in labs, a major challenge for applying it to battery packs is the inconsistency between cells [7], which can cause differences in the boundary values and affect battery safety [6]. This problem also influences the decision-making of the EV regarding when to start preheating and the best preheating temperature to reduce the total charging time, as charging at low temperature can generate large amounts of heat

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that also warm up the battery [8].

This study focused on a high-power electric motorcycle with a battery pack comprising 96 nickel-manganese-cobalt oxide (NMC) prismatic cells in series, each with a rated capacity of 51 Ah. The battery pack provides enough power for a 100 kW motor and a driving range of 350 km. The low temperature range is defined as 0–15 °C considering the marketing by the company. The main demand for this motorcycle is a fast charging speed in this temperature range. The objective of this study is to map the best preheating temperature under different conditions. Battery cell and pack tests are performed to establish a model for obtaining the best preheating temperature. Cell tests are performed to define the safety boundary. Simulations and experiments are performed to test different preheating temperature patterns and determine the best preheating performance.

The rest of the paper is organized as follows. Section 2 presents experiments on the heat generation and dissipation performances of battery cells and the battery pack. The safety boundaries of the bidirectional pulse and fast charging current are tested by a reference electrode (RE). Section 3 presents a zero-dimensional battery cell model as the basic unit and a simplified thermoelectric battery pack model for onboard application to consider the inconsistency between cells. Section 4 presents the experimental verification of the simplified model, which is then used to analyze the inconsistency of the battery pack and map the best preheating temperature in different circumstances. The performance of the proposed preheating method is evaluated by comparing the total charging time with the charging time of no preheating. Section 5 presents the main conclusions.

2. Performance experiments and boundary test

Experiments are performed to evaluate the performances of the battery cells and battery pack. The bidirectional pulse heating technology is set to a pulse frequency of 2 Hz, which is the highest steady-state frequency available for the physical bidirectional DC/DC module used in charger that available in the market.

2.1. Cell experiment

2.1.1. Basic test

First, the battery cell capacity is tested at the begin of life (BoL). The cell is set in a 25 °C chamber for at least 4 h and is then charged according to the constant current (CC)-constant voltage (CV) method. The CC is 0.33C, and the CV is 4.2 V with a cutoff current of 0.05C. Then, the cell is discharged with a current of 0.33C and cutoff voltage of 2.8 V, after 30 min of rest. The test is repeated three times to obtain the average capacity. Next, the open-circuit voltage (OCV) curve is measured in increments of 5 % for the state of charge (SOC) at 25 °C; it is assumed the same curve at different temperatures in this study. To build a precise model that could simulate different behaviors of the battery cell under pulse excitation and continuous charging conditions, the resistance is measured with two different methods. To simulate the battery behavior under pulse excitation, the Ohm resistance is measured according to the hybrid pulse power characteristic (HPPC) test at different SOC and temperatures. The HPPC test applied a pulse of 1.5C with a period of 0.5 s and pause of 30 s. To simulate the behavior under fast charging conditions, the continuous charging resistance is measured at different charging currents and temperatures by using the CC-CV method with CC values of 0.5C, 1C, 1.5C, 1.5C, and 1C and temperature values of 0 °C, 10 °C, 25 °C, 35 °C, 45 °C, respectively. The entropy potential is measured by testing the OCV at temperatures of –20 °C, 0 °C, 25 °C, and 45 °C in intervals of 20 % SOC.

2.1.2. Boundary test

When charging at low temperatures, LIBs suffer from lithium plating, which causes safety issues [9]. For commercial application of bidirectional pulse excitation heating and fast charging technology, the

boundary of the lithium-plating-free charging current must be determined. Previous studies determined the maximum bidirectional current by testing the state of health after preheating [5,10], but a full-scale endurance test consumes a huge amount of resources, time and labor. In this study, conservative boundary for the bidirectional pulse amplitude and fast charging current are tested by a RE implanted in a pouch cell which made from the same material of the 51 Ah prismatic BoL cell. Then the safety boundary is transferred to the big format cell with its C-rate [11]. REs are used to test the potential of the anode, which facilitates a faster development process and can avoid lithium plating. Then the safe boundary of the pouch cell is scaled with a factor to fit the prismatic cell. To define the safe boundary of an aged cell, the same test method can also be used in future work. Previous experiments [5,10] showed that LIBs can be heated by AC or DC pulse greater than the lithium-plating-free boundary, and the battery health will not be damaged irreversibly even if the AC or DC pulse causes the LIB to go over the cut-off voltage. This means that heating with the lithium-plating-free boundary tested by the RE is a conservative method, but not as fast as heating with the boundaries that obtained by other experimental methods.

In this study, REs are made of copper wire. Because it is difficult and unsafe to insert the copper wire (50 μ m to reduce the bias potential [12]) into the prismatic cell, a small pouch cell of 1.6 Ah is made with the electrode and electrolyte materials dissembled from the prismatic cell in a dry room [13], as shown in Fig. 1. To assemble the pouch cell, a copper wire covered with a small piece of separator is inserted between two electrodes, also shown in Fig. 1. After the pouch cell assembled, a clamp is used to vent the gas and compress the cell. The capacity of the pouch cell is tested by the same method of the prismatic cell. The tested capacity remained constant for three cycles indicates that the pouch cell is made successfully. To test the safe boundary value of the bidirectional pulse at different SOC and temperatures, a series of pulse charging currents with different amplitudes are applied to the pouch cell. The pulse duration is 1 s, and the amplitude is from 2 A to 7 A. The anode potential is obtained by the RE. To obtain the boundary value of the fast charging current, the negative potential measured by RE is controlled to 10 mV above zero by the CV mode during charging.

2.1.3. Heat generation and dissipation test

To obtain a precise thermal model of the battery cell, the basic thermal coefficients including the thermal capacity and thermal conductivity are obtained from the battery cell manufacturer. Then, a heat generation and cooling test is performed to verify the model. Fig. 1 shows the experimental setup for the battery cells. Each cell is placed in a chamber and connected with a charger. The thermocouples are connected to a data acquisition equipment, to collect the changes in temperature. The 51 Ah prismatic cell is packaged with two laminations. To obtain the core temperature of the cell, thermocouple 1 is inserted into the middle of the two laminations after a hole is drilled into the side of the shell. Thermocouple 2 is stuck on the surface of the cell to obtain the temperature of the shell. The battery cell is heated by 1C and 2C bidirectional pulse current with 2 Hz frequency, under the temperature of –20 °C and 0 °C with 30 % SOC, after which it is cooled for some time.

2.2. Battery pack experiment

To build an accurate thermal model, experimental data on the battery pack needed to be obtained. The battery pack comprised 96 cells in series and packed into eight modules each containing 12 cells. As shown in Fig. 2(a). For the cooling test, the pack is left at room temperature and then placed in the environment chamber until the cell temperature reaches the set chamber temperature. For the heat generation test, the battery pack is heated by 2C bidirectional pulse current in the environment chamber at –10 °C until the highest cell temperature reaches 25 °C. Then, the pack is charged by multi-stage constant-current method provided by cell manufacture. The cell temperature is measured by

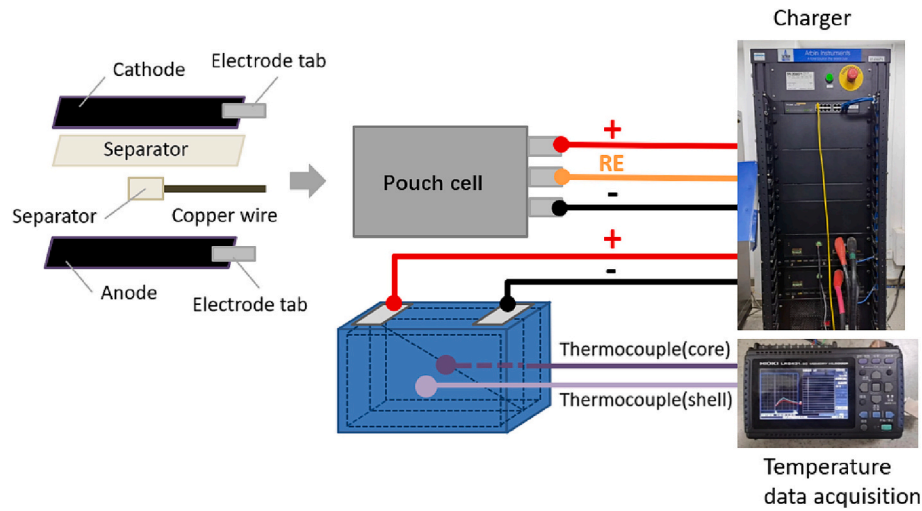


Fig. 1. Experimental setup for the battery cells.

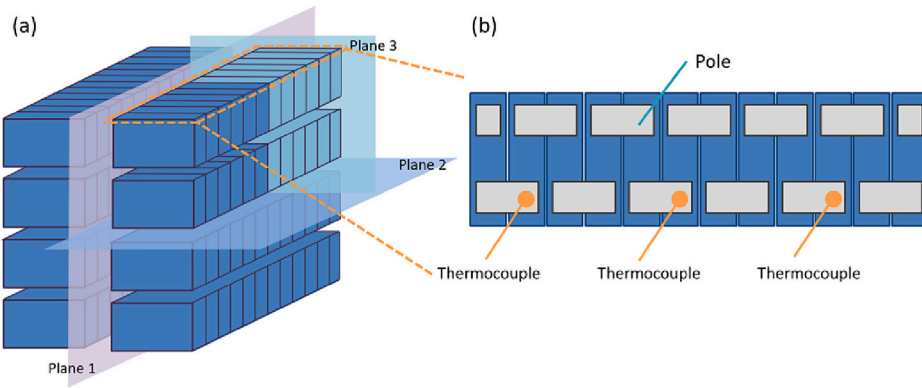


Fig. 2. Structural analysis of the battery pack: (a) symmetry plane and (b) thermocouple locations.

thermocouples inside the battery pack, which stuck on the positive or negative poles of the cells, as shown in Fig. 2(b). The temperature data is transferred through the Controller Area Network (CAN) bus of the battery management system. Although the poles experience a greater evolution in temperature than the rest of the cell because of their large

thermal conductivities and small thermal capacities [14], the temperature at the measurement point is adequately representative of the average cell temperature for the same reason.

The outdoor heat generation and cooling tests of the battery pack are performed in the outdoor environment of Beijing in winter, for which

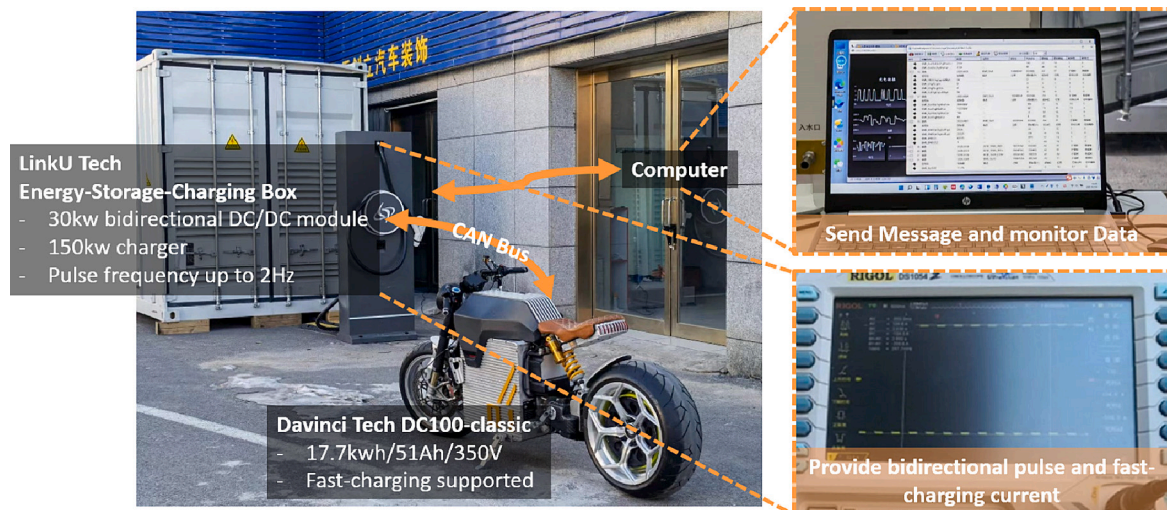


Fig. 3. Outdoor experimental setup: the electric motorcycle (DC100-classic, Davinci Tech) is preheated and charged by an energy storage-charging box (by LinkU Tech).

the outdoor temperature is $-5\text{ }^{\circ}\text{C}$ to $7\text{ }^{\circ}\text{C}$. The experimental setup is shown in Fig. 3. An energy storage–charging box is used to provide the bidirectional pulse. The box is provided by Beijing LinkU Technology Co., Ltd., and it contains a 30 kW bidirectional DC/DC module. The bidirectional module allowed the 150 kW charger to provide a stable DC pulse with a maximum pulse period of 460 ms. The electric motorcycle equipped with a 17.7 kWh battery pack is provided by Davinci Technology Co., Ltd. The motorcycle is connected to the charger through a regular charging gun, and it communicated through CAN bus. The energy storage–charging box is controlled by a computer during tests. To ensure that the bidirectional pulse is the same as the computer command, an oscilloscope is used to observe the current waveform on the output wire of the charger. In this test, the pack is firstly heated with a bidirectional pulse of 100 A and 2 Hz, and then cooling for one day.

3. Battery pack model

The battery cell is modeled firstly, after which the battery pack is described as a lumped parameter model.

3.1. Cell model

Among thermal models, the lumped parameter model ignores the non-uniformity of battery cells and has high computational efficiency [15]. In this study, the battery cell is represented by a two-state thermal model [16], as shown in Fig. 4. The cell can be abstracted into two mass points: the core and shell. The heat is generated in the core, and then transferred to the shell, finally dissipates from the shell by convection cooling. Both the heat transfer and dissipation processes can be described by the thermal resistance model [17] if it is assumed that the shell and core temperatures are both uniform and the heat generation is uniformly distributed within the cell [18]. Simulations by Liu et al. [18] showed that the model realized root mean square error (RMSE) values of $0.82\text{ }^{\circ}\text{C}$ and $0.69\text{ }^{\circ}\text{C}$ for the core and shell temperatures, respectively.

The heat generation and dissipation of the cell follows the Bernardi formula [19]:

$$Q_{bat} = I^2 R - IT_{core} \frac{dOCV}{dt} \quad (1)$$

where I is the current of the cell, R is the resistance of the cell, T_{core} is the average temperature of the core, and $\frac{dOCV}{dt}$ is the entropy potential coefficient. The heat transfer from the core to the shell can be expressed as follows:

$$M_{core} C_{core} \frac{dT_{core}}{dt} = Q_{bat} - \frac{T_{core} - T_{shell}}{R_{core2shell}} \quad (2)$$

where M_{core} is the mass of the laminations, C_{core} is the thermal capacity of the laminations, T_{shell} is the average shell temperature of the shell, and $R_{core2shell}$ is the thermal resistance between the core and shell.

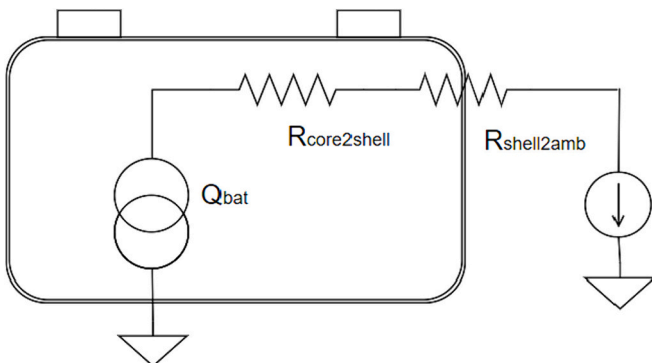


Fig. 4. Two-state thermal model.

$R_{core2shell}$ is calculated from the components in the x , y , and z dimensions, which are given by R_x , R_y , and R_z :

$$\frac{1}{R_{core2shell}} = \frac{1}{R_x} + \frac{1}{R_y} + \frac{1}{R_z} \quad (3)$$

The thermal resistance in three dimensions is calculated as follows:

$$R = \frac{\delta}{\lambda} \quad (4)$$

where δ is the thickness in the x , y , or z dimension and λ is the thermal conductivity in that dimension. Table 1 lists the thermal conductivity and capacity of the core provided by the manufacturer.

The heat dissipation is expressed as follows:

$$M_{shell} C_{shell} \frac{dT_{shell}}{dt} = \frac{T_{core} - T_{shell}}{R_{core2shell}} - \frac{T_{shell} - T_{amb}}{R_{shell2amb}} \quad (5)$$

where M_{shell} is the mass of the shell and C_{shell} is the heat capacity of the shell, which is made of aluminum. $R_{shell2amb}$ is the thermal resistance between the shell and chamber temperature, and it is calculated as follows:

$$R_{shell2amb} = \frac{1}{hA} \quad (6)$$

where h is the convective heat transfer coefficient and A is the total surface area of the cell.

3.2. Battery pack model

According to Fig. 2(a), the modules on the two sides of Plane 1 can be considered the same in terms of heat generation and dissipation. According to the symmetry principle, the battery cells are also symmetrical with Plane 2 and Plane 3 when modeling. Thus, only four thermocouples need to be considered, as shown in Fig. 5(a). Therefore, the battery pack model can be simplified to a zero-dimensional thermal model comprising only 12 cell models, the air in the battery pack, and other components of the battery pack (i.e., case). Each cell is next to another cell, a symmetry plane, or the case. Depending on the material and object at the contact surface, cells are classified into types A–D representing different heat dissipation classes, as shown in Fig. 5(a). Type A has three surfaces in contact with the case and one surface in contact with an adjacent cell. Type B has two surfaces in contact with the case and one surface in contact with an adjacent cell. Types C and D both have two surfaces in contact with adjacent cells, but type C has two surfaces in contact with the case while type D has only one. All other surfaces are considered adiabatic owing to symmetry or the negligible thermal resistance. Fig. 5(b) shows the established lumped parameter thermal model of the battery pack. The cells are described by the two-state thermal model in Section 3.1 but with differing heat dissipation processes according to the type.

4. Results and discussion

4.1. Experimental results

The battery resistance is evaluated under different experimental

Table 1
Thermal conductivity and capacity of the core.

Characteristic	Value	
Thermal conductivity W/(m•K)	λ_x	1.02
	λ_y	19
	λ_z	19
Thermal capacity J/(kg•°C)	1020	

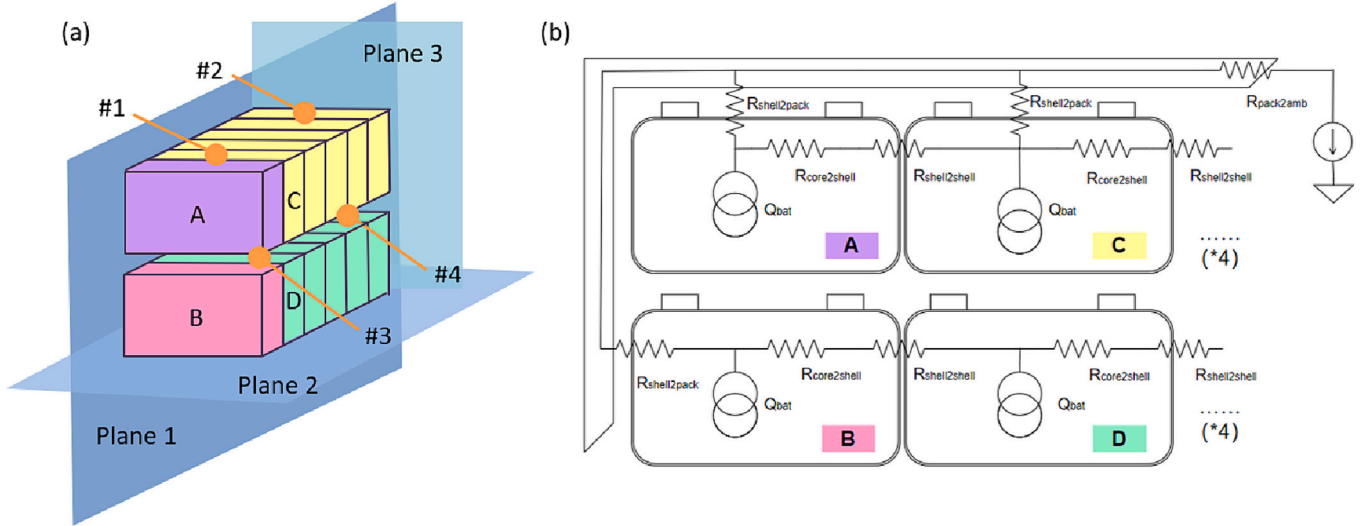


Fig. 5. Battery pack model: (a) four types of cells and thermocouple locations; (b) lumped parameter model.

conditions, shown in Fig. 6 (a)-(c). The z axes represent the charging or discharging resistance calculated from the HPPC test, and the continuous charging resistance respectively. The entropy potential coefficient calculation method is described in Appendix A.

The boundary of bidirectional pulse and fast charging current is determined by the negative potential of the battery while charging, the test results are arranged into maps, as shown in Fig. 6 (d) and (e). In this research, the charging current is considered to be lithium-plating-free if the negative potential is above 10 mV. For the bidirectional pulse boundary, the exact charging pulse which makes the negative potential just reach 10 mV is calculated by the test results described in Section 2.1.2. Two current pulses are extracted from the series of charging pulses with different amplitudes: the maximum pulse I_{above} that allows the negative potential E_{above} above 10 mV, and the minimum pulse

I_{below} that allows the negative potential E_{below} below 10 mV. The average resistance R_{neg} of the anode is calculated from these two sets of pulses and the negative potentials as follows:

$$R_{neg} = \frac{1}{2} \left(\frac{E_{neg} - E_{above}}{I_{above}} + \frac{E_{neg} - E_{below}}{I_{below}} \right) \quad (7)$$

where E_{neg} is the equilibrium negative potential after long time resting of the battery. Then, the resistance is used to calculate the pulse amplitude allowing the negative potential to just reach 10 mV. Finally, the boundary value of the maximum pulse current $I_{boundary}$ is calculated as follows:

$$I_{boundary} = \frac{E_{neg} - 10mV}{R_{neg}} \quad (8)$$

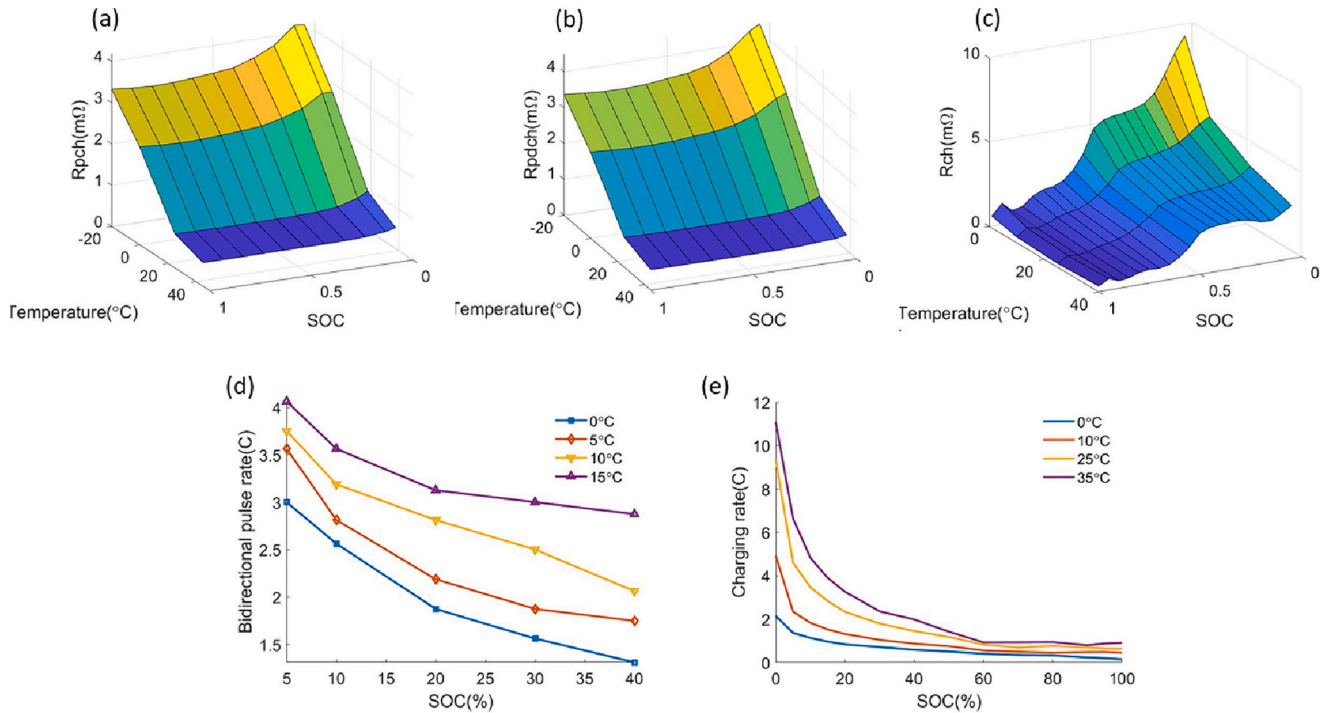


Fig. 6. Parameter identification: (a) 0.5 s resistance during charging, (b) 0.5 s resistance during discharging, (c) continuous charging resistance, (d) bidirectional pulse boundary value, and (e) fast charging boundary value.

This can converted the boundary current into C rate, which will be scaled while applying to the prismatic cell. The fast charging current is determined automatically by the CV method of the charger, as mentioned in Section 2.1.2, to control the negative potential in the lithium-plating-free range. The fast charging current is also converted to C rate, then implements to the prismatic cell.

4.2. Model verification

4.2.1. Cell model

The two-state model is verified by heat generation and dissipation test in Section 2.1.3. The temperature of the core is measured by the thermocouple inside the battery cell, which is tested only once for safety reason, and the temperature of the shell is measured by the thermocouple on the surface of the cell. The results showed that the model accurately simulated the temperatures of both the core and shell. The convective heat transfer coefficient is adjusted to $35 \text{ W}/(\text{m}^2 \cdot \text{K})$ to fit the experimental results [20]. The average RMSE of the two-state model is only 0.17°C (Fig. 7).

4.2.2. Battery pack model

The battery pack model is verified against the results of three experiments. First, the cooling test is used to finely adjust the thermal resistance between the cells and battery pack. The convective heat transfer coefficient (h) of the chamber is $35 \text{ W}/(\text{m}^2 \cdot \text{K})$. Fig. 8 compares the experimental data and simulation results. The RMSE is not considered because the measurement accuracy of the thermocouple is only 1°C (i.e., RMSE must be greater than 1°C). Second, the model is verified against the experimental results for preheating and charging under different conditions. The chamber had $h = 6 \text{ W}/(\text{m}^2 \cdot \text{K})$, which is the same as the outdoor condition. The model accurately simulated the temperature of the four measurement points during preheating and charging. The charging current could also be simulated by consulting a lookup table, where the charging current is determined by the temperature and SOC of the battery pack. Third, the model is verified for the preheating and cooling process under outdoor conditions as it described in Section 2.2. The heating and cooling data are obtained by the CAN bus of the electric motorcycle, for which only the highest (#4) and lowest (#1) temperatures are recorded. The simulation errors can be partly attributed to fluctuations in the outdoor temperature, which are about 10°C during the test.

4.3. Temperature inconsistency analysis

Inconsistency in temperature leads to inconsistency in the internal resistance, which causes differences in the boundary values for the bidirectional pulse and fast charging current between cells. Thus,

analyzing the temperature inconsistency of the battery pack is important for identifying the cell that determines the boundary value. Fig. 9 shows the heat maps of the battery pack after preheating to 15°C or charging to 15°C . The battery pack is initially at 5% SOC, and the ambient temperature is 0°C . In the preheating process, the battery pack is heated by a constant 2C bidirectional pulse to 15°C . The temperature difference between cells is less than 0.7°C in the shell and 0.2°C in the core. In the charging process, the battery pack is charged to 100% SOC, and the terminal average temperature is 15°C . The temperature difference is 2.7°C in the shell and 2.5°C in the core, which is much greater than bidirectional pulse heating process. The minimum cell temperature is in the corner of the battery pack and thus should be used to determine the bidirectional pulse amplitude, which increases with temperature.

The simulation results indicate that bidirectional pulse heating is an excellent preheating method with a homogeneous performance. Because the cells are all in series, cells with a lower temperature have greater resistance, which generate more heat to raise its temperature. Fig. 10 shows some simulation results illustrating the effects of temperature differences. The corner cell is preheated to 25°C with the bidirectional pulse at the boundary value. The greatest temperature inconsistency appeared under the initial conditions of 40% SOC and 0°C , which requires the longest time to preheat to 25°C . Regardless of the conditions, the temperature difference is less than 0.26°C between cores, less than 0.98°C between shells, and less than 1.43°C between the case and ambient temperature. Therefore, the temperature inconsistency of the cells after bidirectional pulse preheating is assumed negligible. This means that the initial temperature of the charging process can be considered the minimum terminal preheating temperature, and the initial temperature of the case can be considered the ambient temperature.

4.4. Best preheating temperature and performance analysis

The best preheating temperature is defined as the terminal temperature of the preheating process that minimized the total charging time (i.e., sum of the preheating time and charging time). The main objective of this study is to determine the best preheating temperatures under different circumstances so that the EV can decide the preheating temperature on its own. The best preheating temperature can be mapped by the relationship between the initial battery pack temperature (assuming that it is the same as the ambient temperature) and initial SOC. The temperature and SOC determine the battery pack resistance together, which determines the temperature increase rate of the preheating process. Thus, they also determine the charging time. In addition, the charging capacity can determine the best preheating temperature, with the same optimization methods as the one used for charging to 100% SOC. For further application, the minimum charging time can be

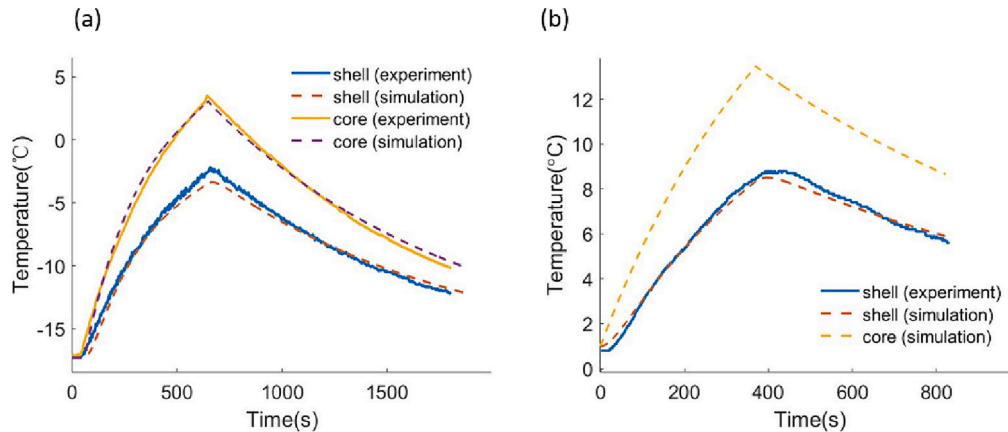


Fig. 7. Verification of the two-state thermal model: (a) heated with 1C 2 Hz pulse current; (b) heated with 2C 2 Hz current.

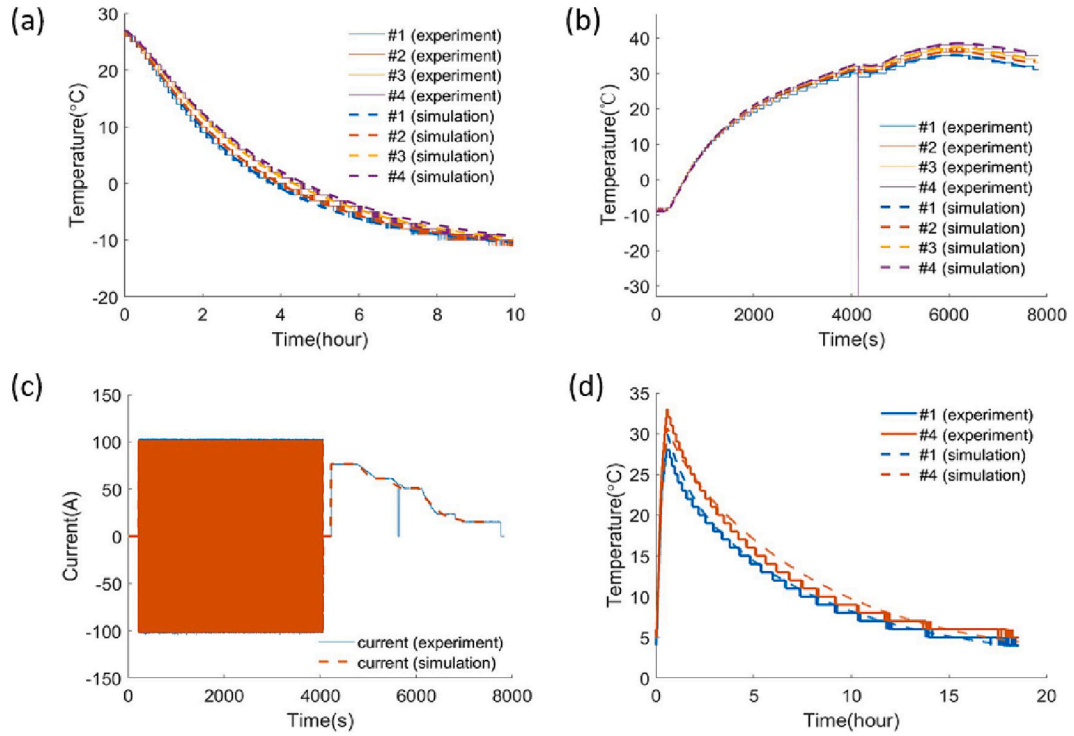


Fig. 8. Verification of the battery pack model by comparison between the simulation and experimental results: (a) temperature when cooling in a $-10\text{ }^{\circ}\text{C}$ chamber; (b) temperature and (c) current when preheating in a $-10\text{ }^{\circ}\text{C}$ chamber before charging; (d) temperature when preheating and cooling outdoors during winter in Beijing.

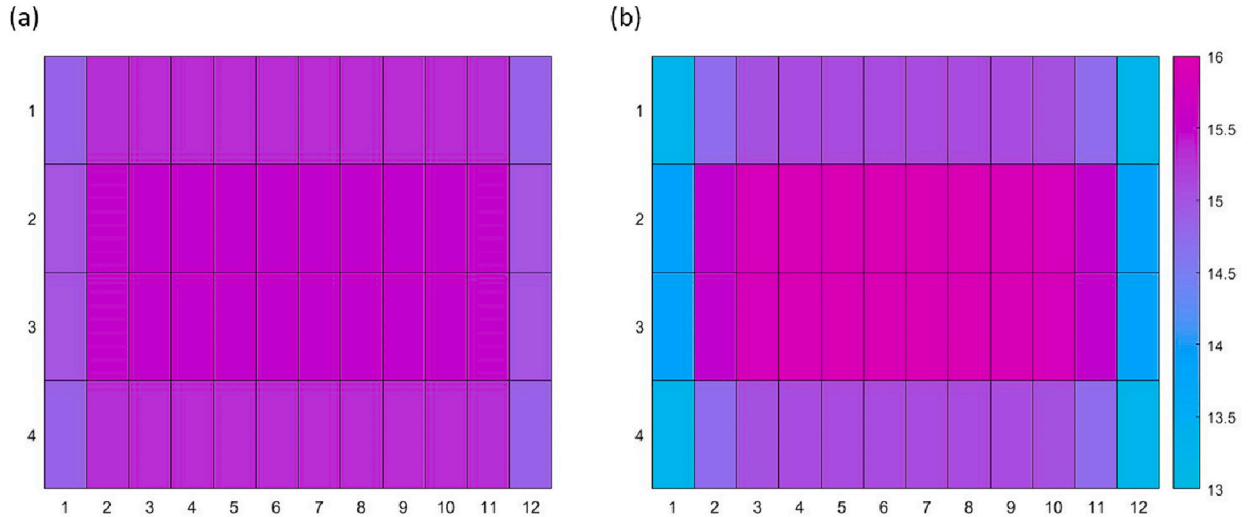


Fig. 9. Temperature inconsistency of the battery pack: (a) temperature distribution for preheating to $15\text{ }^{\circ}\text{C}$ under the initial conditions of 5 % SOC and $0\text{ }^{\circ}\text{C}$; (b) temperature distribution for fast charging to 100 % SOC under the initial conditions of 5 % SOC and $0\text{ }^{\circ}\text{C}$.

calculated on board.

To search the best preheating temperature, an optimization method that decouples the preheating and charging processes is used. The optimization target is the shortest charging time, not considering the energy consumption [21], because the heating energy comes from an external power source (energy storage-charger box), and the bidirectional pulse heating has a high efficiency. Firstly, simulations are performed of preheating the cells to $25\text{ }^{\circ}\text{C}$ at different combinations of the SOC and ambient temperature. The results indicate that the preheating time could be approximated as a cubic function of the cell temperature ($f1$ in Fig. 11), where the preheating temperature only increases with the increasing of the heating time. The RMSE of this fitting is only 0.1449 s.

Based on the assumption in Section 4.3, the cell temperatures are considered the same in the battery pack at the end of the preheating process. Thus, the initial condition for the charging process is easy to obtain. The initial temperature of the charging process is the terminal temperature of preheating. Simulations are performed of the charging process at different preheating temperatures and use different combinations of the SOC and ambient temperature. The results indicated that the charging time could be approximated as a quadratic function ($f2$ in Fig. 11), where the RMSE of the fitting is 11.65 s. The reason why the charging time start to increase after certain preheating temperature is that the cell supplier does not allow the cell to charge after $55\text{ }^{\circ}\text{C}$, the charging current must decrease after $30\text{ }^{\circ}\text{C}$ as the pack is air cooled. So

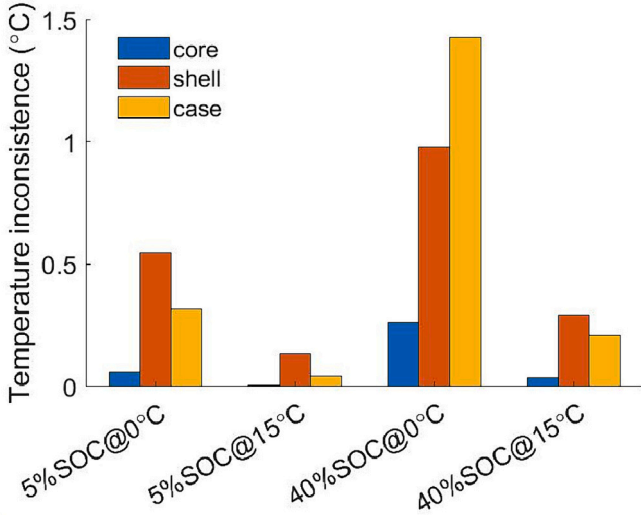


Fig. 10. Temperature inconsistency for preheating to 25 °C.

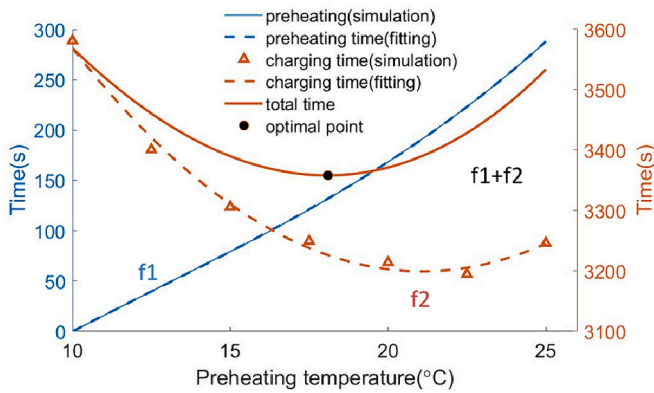


Fig. 11. Example search of the best preheating temperature under the initial conditions of 5 % SOC and 10 °C.

after the initial temperature reaching 25 °C, the charging speed at the end of charging state is limited by the battery temperature. Thus, the total charging time ($f1 + f2$) is the sum of the preheating time($f1$) and charging time($f2$) (red line in Fig. 11). The best preheating temperature is given by the x-axis value at the curve minimum (black mark in Fig. 11), and the shortest total charging time is the corresponding y-axis value.

The law of the best preheating temperature is mapped in Fig. 12(a).

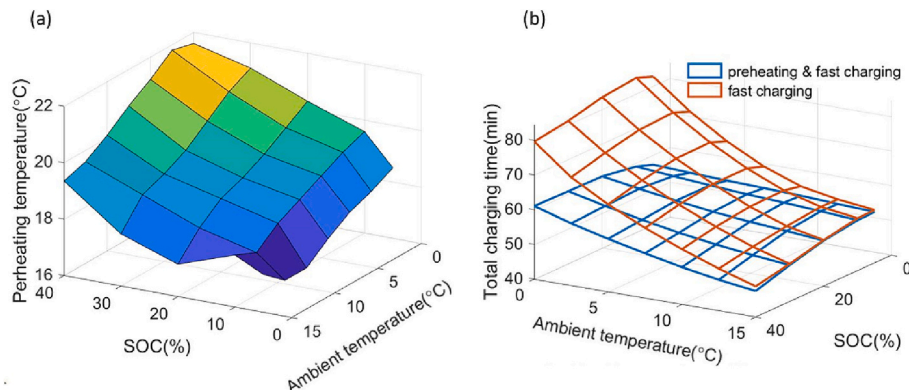


Fig. 12. (a) Best preheating temperature map; (b) comparison of total charging times with and without preheating.

At a high SOC, the preheating temperature tends to be higher because both the charging current and internal resistance are lower than at a low SOC, so the charging process does not generate a significant amount of heat. When the initial temperature is low, the internal resistance is high, but the charging current is limited by the temperature, so a preheating process is still necessary to reduce the total charging time. Fig. 12(b) compares the total charging times with and without preheating. Preheating increases the charging speed significantly at low temperatures, but the benefit fades as the ambient temperature increases. When the ambient temperature reaches 15 °C, the charging speeds with and without preheating are nearly the same. Thus, preheating the battery pack is unnecessary when the ambient temperature is over 15 °C.

Fig. 13 shows some example applications of the best preheating map. First, the ambient temperature and initial SOC are set to initialize the model. Then, the model searches for the bidirectional pulse based on the cell temperature and SOC, and the battery pack is heated to the best preheating temperature. After the minimum cell temperature reaches the specified value, the model switches from preheating to fast charging. The model also searches for the charging current based on the cell temperature and SOC. During the preheating and fast charging processes, the thermal model of the battery pack calculates each temperature of the cells. Fig. 13(a) shows the charging process at a low SOC and low temperature of 5 % SOC and 0 °C, respectively. The time when the preheating switches to fast charging is indicated by the orange dashed line. A simulation is performed comparing the current, SOC, and minimum core temperature among cells with and without preheating. The preheating time is as short as 164 s because the average temperature increase rate reached 6.8 °C/min. This is also the highest temperature increase rate of the battery pack because the internal resistance is greatest under these conditions. Fig. 13(b) shows the charging process under the initial conditions of 40 % SOC and 0 °C, whose preheating time is longer but also shortens the total charging time.

Table 2 presents the performance of the bidirectional pulse preheating by simulation. The reduction percentage is the difference between the fast charging time without preheating and the total charging time with fast charging and preheating, divided by fast charging time without preheating. The best performance is obtained at the initial conditions of 5 % SOC and 0 °C, where the proposed method reduced the total charging time by 31 %.

The motorcycle currently uses a multi-stage constant-current charging method, which is slower than the fast charging method based on RE in this research. Table 3 presents the performance of using both bidirectional pulse preheating and fast charging technology by simulation. The reduction percentage is the difference between the multi-stage constant-current charging time and the time of the proposed method, divided by the multi-stage constant-current charging time. The best performance is also obtained at the initial conditions of 5 % SOC and 0 °C, where the proposed method reduced the total charging time by 74 %.

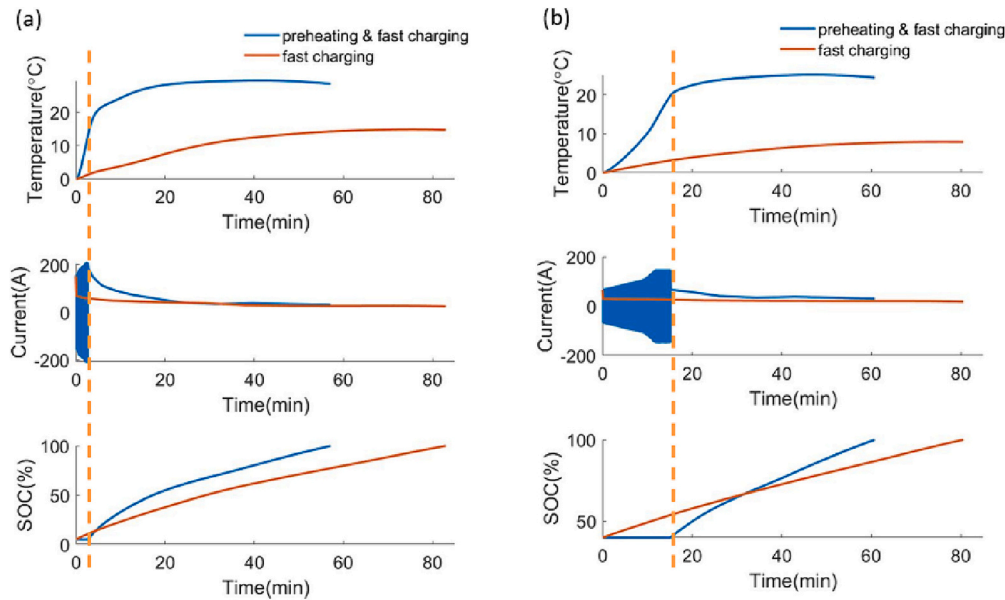


Fig. 13. Comparison of charging processes with and without preheating: (a) 5 % SOC and 0 °C; (b) 40 % SOC and 0 °C.

Table 2

Reduction in charging time by using bidirectional pulse preheating.

SOC	0 °C	5 °C	10 °C	15 °C
5 %	31 %	17 %	4.6 %	1.0 %
10 %	30 %	18 %	5.9 %	1.0 %
20 %	27 %	17 %	7.4 %	0.34 %
30 %	25 %	17 %	8.8 %	0.74 %
40 %	23 %	15 %	9.3 %	2.8 %

Table 3

Reduction in charging time by using bidirectional pulse preheating and fast charging technology.

SOC	0 °C	5 °C	10 °C	15 °C
5 %	74 %	54 %	42 %	28 %
10 %	70 %	52 %	39 %	25 %
20 %	65 %	47 %	35 %	20 %
30 %	63 %	46 %	33 %	20 %
40 %	63 %	47 %	31 %	23 %

5. Conclusions

In this study, a bidirectional pulse heating and fast charging method is developed for application to high-power electric motorcycles, which is also the first tryout for vehicle-used bidirectional pulse heating. The electrical and thermal characteristics of the battery cells and battery pack are tested, and an RE is used to obtain the boundary values for the bidirectional pulse and fast charging current. Then, a simplified thermoelectric coupled battery pack model is established. Finally, the temperature inconsistency of the preheating is analyzed. The results demonstrated that bidirectional pulse heating provides a uniform temperature increase and negligible temperature difference inside the battery pack. The model is then used to generate a best preheating temperature map. The simulation results show that the battery pack should be preheated when the ambient temperature is below 15 °C to reduce the charging time. At 0 °C, the bidirectional pulse preheating showed good performance because of the high internal resistance at low

temperatures. When the initial SOC is 5 %, the temperature increase rate reached 6.8 °C/min with the safety boundary value of the bidirectional pulse, which reduced the charging time by 74 % comparing to the current charging time. The bidirectional pulse heating and fast charging technology will be applied to this electric motorcycle in the future. Also, the aging behavior under bidirectional heated-charging will be carried out after the implementation of the method.

CRediT authorship contribution statement

Meilin Han: Writing – original draft, Validation, Software, Methodology, Data curation. **Languang Lu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Yalun Li:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Shuoyuan Mao:** Writing – review & editing, Methodology. **Yufang Lu:** Writing – review & editing, Supervision. **Yudi Qin:** Methodology, Conceptualization. **Hewu Wang:** Writing – review & editing, Supervision. **Huaishi Sha:** Data curation. **Zhi Yao:** Data curation. **Yong Lu:** Writing – review & editing, Supervision. **Jingwei Wang:** Writing – review & editing, Conceptualization. **Minggao Ouyang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Calculation of the entropy potential coefficient

If the entropy potential coefficient is assumed constant at different temperatures but is related with the SOC [22], then it can be calculated by linear regression:

$$\frac{dOCV}{dT} \approx \frac{\Delta OCV}{\Delta T} \quad (a1)$$

Figure a1(a) shows the linear regression results with the least-squares method at different SOC. Fitting lines are given for 20 %, 50 %, and 100 % SOC. Figure a1(b) shows the calculation results for the entropy potential coefficient. The coefficient is less than zero at both low and high SOC but is above zero in the middle.

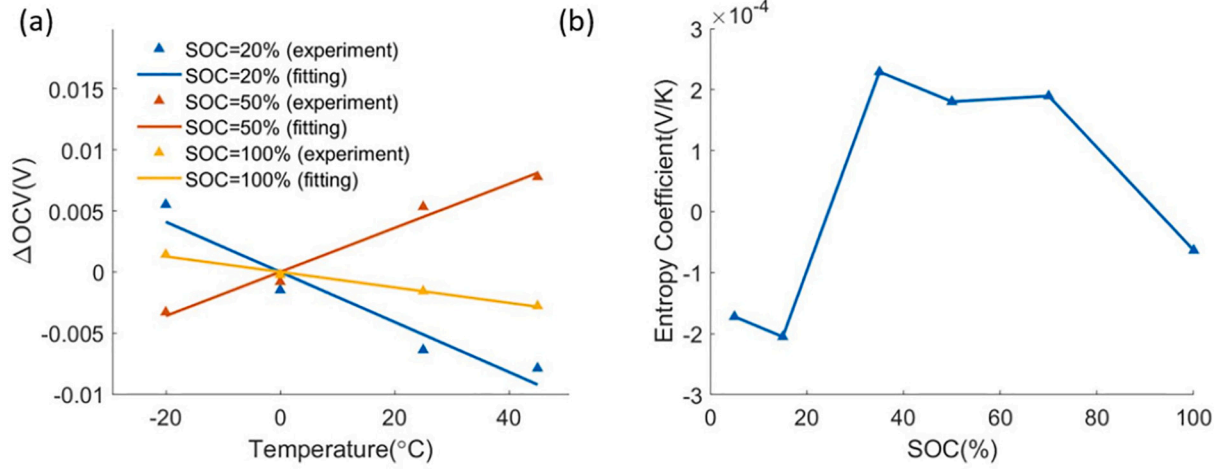


Figure a1. Entropy potential coefficient: (a) identification by linear regression; (b) calculated entropy potential coefficients.

Appendix B. Calculation of the thermal resistance

A heat transfer test is needed to determine the thermal resistance between two objects. However, once the battery pack is assembled, it is not easy to take out a battery module. Thus, the thermal resistance between the modules and case is difficult to test in normal experiments. This parameter is usually determined empirically or is adjusted manually. To obtain a more precise model, a commercial physical model is used as a reference. When developing a battery pack system, the manufacturer always builds a three-dimensional thermal model based on the material characteristics, and the model is calibrated against experimental data. In this study, a three-dimensional thermal model is built in *Fluent*® to obtain precise parameters for the simplified battery pack thermal model.

As shown in Figure b1(a), the thermal resistance between adjacent cells is calculated by setting the orange cell temperature to 40 $^{\circ}C$ and the blue cell temperature to 0 $^{\circ}C$. The orange cell is kept at a constant temperature, and both cells are adiabatic except for the contact surface and material stuck in the middle. As shown in Figure b1(c), the thermal resistance between the cells and case is calculated by setting the cells to a constant temperature of 40 $^{\circ}C$, and all cells are regarded as a solid object. The outside surface of the battery pack is set as adiabatic. The thermal resistance between cells and between the cells and case can be calculated as follows [6]:

$$T = T_{\infty} + (T_0 - T_{\infty})e^{-kt} \quad (b1)$$

where

$$k = \frac{1/R_{thermal}}{C_p M} \quad (b2)$$

$R_{thermal}$ is the thermal resistance between objects, C_p is the thermal capacity of an object, and M is the mass of the object. The parameter identification is performed by using *cftool* in MATLAB. The fitting curves are shown in Figures b1(b) and (d).

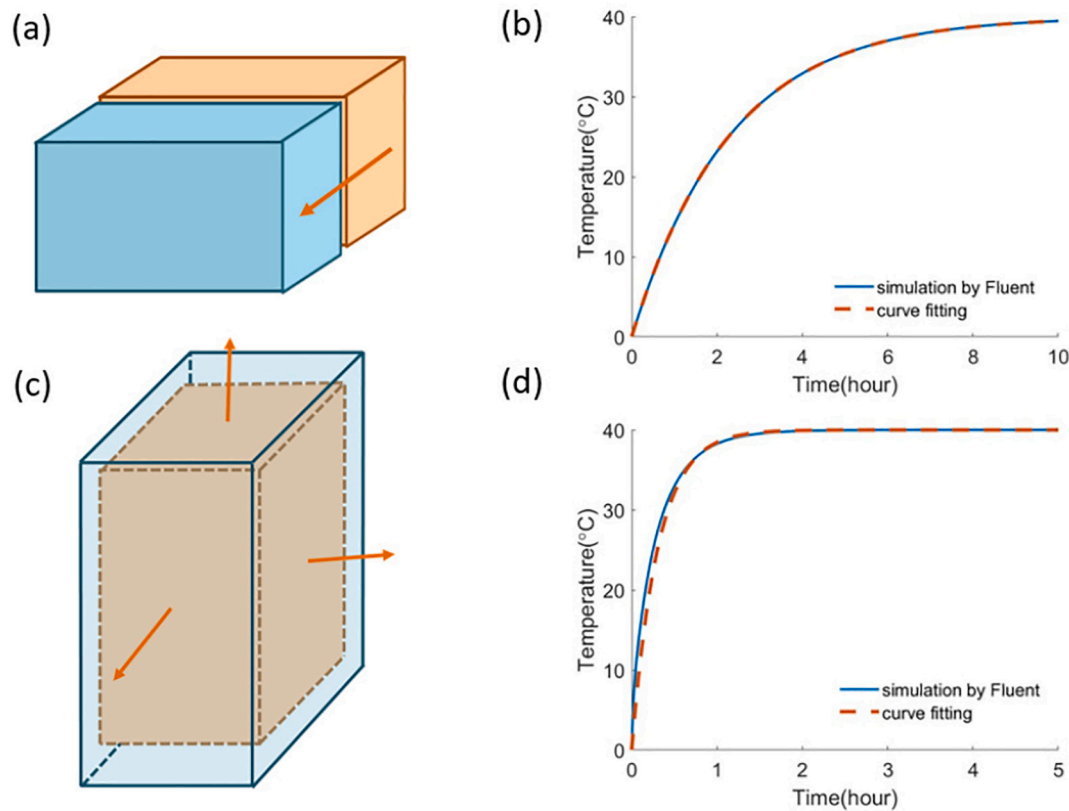


Figure b1. Thermal resistance identification: (a) heat transfer and (b) curve fitting results for the thermal resistance between two cells; (c) heat transfer and (d) curve fitting results for the thermal resistance between the cells and case.

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